



Fig. 6: Peripherals connected to the Raspberry Pi board



Fig. 7: GNOME screen

Setting up the board with an operating system

Preparing the SD card. The first interesting task is to prepare the SD card for a Raspberry operating system. What you need here is a computer with a good Internet connection. A 256kbps or 512kbps broadband connection is just fine.

Download the image of the operating system from 'http://www.raspberrypi.org/downloads.' You can download any of the below-mentioned images to get started:

1. Raspberry 'wheezy'
2. Soft-float Debian 'wheezy'
3. Archlinux ARM
4. RISC OS

After downloading the zip file, extract the image in a suitable folder. Insert the 2GB or 4GB SD card in a card reader and connect it to the computer where you have downloaded the image. At this stage, before moving further, determine what identity is

given to the SD card by your computer—sda, sdb, sdc etc. Any mistake here will lead to the complete data loss from your computer's hard disk.

Linux user. Just after inserting the card, issue below-mentioned command at the terminal prompt to see what name has been given to the card (look for the highest order like sdc and sdd):

```
ls /dev/sd*
```

The result will be something like:

```
efy@Acer-A0D255E:~$ ls /dev/sd*
/dev/sda /dev/sda2 /dev/sda4 /dev/
sda6
/dev/sda1 /dev/sda3 /dev/sda5 /dev/
sda7 /dev/sdb1 /dev/sdb2 /dev/sdc
```

SD card has been identified as '/dev/sdc' in the case mentioned above.

To double-ensure, just unplug and re-plug the card a few times and notice the changes. Now use 'cd' command and go to the folder where you have extracted the image: 'Desktop/Rasperry-PI/OS' is the directory where we extracted the image.

```
cd Desktop/Rasperry-PI/OS
```

This image file now needs to be written on the SD card to make it the OS. At the terminal prompt, issue the following command:

```
sudo dd if=2012-09-18-wheezy-
raspbian.img of=/dev/sdc bs=4M
```

Where 2012-09-18-wheezy-raspian.img is the name of the image. /dev/sdc is the memory card where we want to put this image.

(When it prompts for 'su' password, supply the password.)

This will use 'dd'—the powerful image creator program of Linux—and transfer the image to the SD card. Other than wheezy, archlinux will also get installed the same way. However, for xbmc, the Internet connection is required while transferring the image onto the SD card.

Windows user. Use 'image write' program downloaded from 'https://launchpad.net/win32-imager-writer.' Run the .exe file. You will see a window

as shown in Fig. 4. Browse to locate the path of the image file. Take care that you select the right device, otherwise it may format your hard drive.

Once the write is successful, you can immediately put the SD card in your Raspberry Pi and boot it for the first time.

First boot. Insert the SD card on the back side of the Raspberry board, connect the cables, keyboard, mouse and power supply. A login prompt appears after a handful of lines.

The default login is "Username: pi, Password: raspberry."

By default, Raspberry Pi does not start the GUI mode. So type 'startx' at the prompt and the beautiful GNOME desktop unfurls on the monitor screen. A TV screen output certainly lacks clarity but on an HD TV it is very good. Generally, the display works for most of the TVs and the HDMI displays but occasionally it goes awry and the picture becomes snow-like—very large or very small. For that you need to change the resolution setting of the displays.

Space and time. Play with icons on the desktop. Right-click, left-click, explore the Raspberry Pi your way. Go to a terminal window and change the date by the following command:

```
pi@raspberrypi:
/$ sudo date -
set= 2013-02-02
23:30
```

This is necessary as some of the programs will not install if the date is not correct.

Raspberry team has not included a real-time clock on board to keep the cost within limit. Therefore every time it reboots, the time changes back.

Raspi-config. To do the basic first-boot configurations, open 'LXTerminal' and type 'sudo raspi-config.' You will see 'raspi-config' utility. This utility also appears when you boot the system for the first time.

Following are the Raspberry Pi configurations:

expand-rootfs. This option expands the root file system to fill the SD card. Bring the cursor on it and then press 'Enter.' Don't worry, the Raspberry Pi